

# CampusFix -Smart Campus Complaint Management System using Machine Learning

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**Abstract-** It is common to find in educational institutions a lag in responding to the issues raised by students on infrastructural facilities, repairs, as well as university support services. The conventional methods of complaint handling are based on manual methods, which causes confusion, lack of transparency and slow response. CampusFix which is a smarter strategy of managing complaints in a campus through the use of the AI matrix that does the sorting, prioritizing, assignment and a forecast of wait time of a problem is provided below. It has in development an app built with React Native to allow students to easily report things besides, a powerful FastAPI engine that runs numerous clever models to analyze text, understand photos as well as make smarter decisions. This flow operates with NLP high-order transformers, basic machine learning algorithms, CNNs to understand images and ensemble learning algorithms to calculate what must be focused on the first. The results of the test reveal that the proposed system demonstrates better accuracy in the categorization of the complaints, the results are more consistent when using multimodal input, and the administrative efficiency in the proposed system is higher. This arrangement therefore offers an informational-based model which is useful towards intelligent campuses or metropolises.

**Keywords:** Multimodal Complaint Management System, Transformer-based Text Classification, Computer Vision-based infrastructure monitoring, Priority prediction, Automated Campus maintenance.

## I. INTRODUCTION

An effective complaint management procedure has the dual role of ensuring the physical environment of the campus buildings is maintained, and that the students are also content. Nevertheless, most schools continue to employ the traditional techniques of using paper forms, talking to employees or sending unstructured emails. Consequently, much time will be wasted in fixing them and certain issues will be lost and no one is actually responsible. Pests, dead Wi-Fi, bad lighting, dirty washrooms these are the things that keep on appearing and ruining life at school again and again. In the absence of a centralized reporting system, the complaint may undergo more time before it is resolved.

Recently, AI and machine learning enabled the implementation of some automatic capabilities in areas such as help desks, city planning and school buildings. The implementation of this technology on the college campuses would make the interaction between the students and staff easier. Intelligent models are able to classify complaints, approximate the speed with which some action should be taken, read hints on photos, guess how much time a fix requires - features which cannot be performed on more traditional tools. However, most of the current campuses continue to operate based on fixed rules that when things keep on changing, they find themselves getting strained.

In order to address these problems, CampusFix presents a multi-input reporting tool that is machine learning-based and designed specifically to be used in schools. It applies text analysis with the help of NLP, visually with CNNs, composite models to predict urgency, and predictions on the time with the help of regression models. The phone-friendly access, the ability to auto-assign reports, and forecast tools all help see that there is in fact a consistent process that is slowly increasing the pace at which problems are processed at the campuses. Rather than individual systems, it combines language processing, image reading, and prediction methods into a single linked system - this makes sense of complaints more accurately and provides guidance on choices using actual data. With this combination, reports are treated more cautiously, solved more effectively, and forwarded directly to the appropriate office, and colleges have an opportunity to improve the manner in which they address concerns of students.

Besides the already in place methods, CampusFix utilizes textual information coupled with image-based input as a way of giving better results. Due to this reason, it is much superior when dealing with unpleasant scenarios such as unclear reports, omissions of information, or distorted images. This does not mean that the system is designed in the ways that it cannot change and that in case the school needs to adopt a new tool tomorrow, say an AI model, or changes in their ways of reporting they will have to start building on the new start. It largely operates on its own and users are presented with what is transpiring behind the scene and assists the staff members in making sound decisions in their daily decisions.

This improvement leads to smoother running of campus, happier students, and improved management team that is able to identify problem areas earlier than later. Such systems can be used to undertake the large-scale civic management.

## II. RELATED WORK

The vision-based infrastructure inspection system that was presented in Lin et al. (2017) targets the detection of objects and defects in diverse sizes in the complex scenarios like buildings and urban areas. Their experiment demonstrated that it is possible to apply automated visual systems to aid in the detection of small-scale and larger scale structural problems in imagery. However, the system was more of image-level detection and did not allow the generation of automated incident report or adding annotations by user complaints, text descriptions, administrative workflows to resolve reported content.[1].

Sandler et al. (2018) introduced a resource-aware VAST tool that was developed on the mobile platform. The system also enabled faster image processing and real time feedback, which may be applicable in a situation where any users want to post images using mobile devices. This was good in processing images on-device, but visual inference was the only thing it could perform and could not be utilized in a complete-fledged complaint handling system - imagine text input processing, classifying the issue and emailing the maintenance personnel. [2]

The scheme of real-time object detection by Redmon et al. (2016) was introduced and has since been widely applied in surveillance, safety monitoring, infrastructure inspection etc. In this respect, they are able to identify harmful or bad things in images or video images very fast. These solutions are, however, personal detection methods and do not assist in submitting complaints or prioritizing them according to their severity as is required by institutionalized complaint management applications [3].

He et al. (2016)-introduce Extra-Large Visual Recognition Model (E-LV-VCR) whose performance is far more effective than the recognition of complicated visual patterns in pictures. In numerous applications, the defect, damage and environmental condition detection have been carried out using these systems. However, it has not thought about the broader application of addressing the user-reported issues, as it combines textual descriptions and visual evidences automatically or offers end-to-end resolution processes on a large-scale campus basis. [4].

Baltrusaitis et al. (2019b) examined the reaction of multi-modal systems where it interacts some textual and visual inputs jointly to facilitate understanding in actual applications of multi-modal systems such as social media analysis, and human-computer interaction. They also emphasized the fact that with incomplete or ambiguous input, a multi-modal system is more flexible. However, the analysis focused on thematic models and theoretical implementations, as opposed to developing a real time, deployed system of management of complaints and scoring, prioritization and administrative decision support. [5].

The models suggested by Wang et al. (2018) were capable of encoding the global contextual information about the visual

data, hence giving a larger vision of the intricate scenes. These systems are applicable when the spatial dependencies and the environmental context are important. Nonetheless, they did not serve the purpose of handling complaints and was not associated with textual reporting, evaluation of severity and automation of management processes in smart campuses systems.[6]

Hu et al. (2018) suggested better visual frameworks, that present key visual information, and hold back non-critical information in favor of enhancing sensitivity to important areas in images. Such systems are effective to locate some significant visuals such as damage/degradation. However, they were only used in visual identification regarding use-cases and no such follow up is made on multimodal complaint analysis, priority prediction or automated issue solve pipelines. [7].

Despite the previous application of object detection and multimodal learning, and the effective deep network structures, the existing research has certain apparent gaps when used in the actual complaint system of the campus.[8] The current systems were either rule-based systems, or multimodal-representation systems based on text-only analysis which could not handle such and was not able to rank effectively attacks to authorities. The CampusFix fills these gaps by the combination of the powerful deep learning frameworks like ResNet, YOLO, MobileNetV2, FPN, Non-Local Networks and SE modules and multimodal fusion mechanisms to address noisy descriptions, ambiguous images, multi scale defects and context sensitive severities. In addition, CampusFix can provide predictive analytics to identify the resolution time and a mobile integration into a real-time platform to overcome the challenge of scalability, transparency, and automation as a hindrance of the earlier solutions. It is a holistic design and thus the problem of fragmentation that existed in earlier designs is handled in the system and provides an efficient and intelligent civil approach to the existing smart campus.

## III. PROPOSED METHODOLOGY

The CampusFix system as a whole is an optimised and multi-layered multi-module system that enables the harmonic interaction of a user with backend services and a number of machine learning components as shown in Figure 1. On the top of I-Complaints architecture are the students and administrative offices as a part of the User Layer interface with the system through the web and mobile interface and all requests are redirected through the Authentication Module and to provide real-time updates on the status of a complaint Notification Service is placed. Such operations are sent to the Application Layer, at which the FastAPI server serves as the main coordinator when it comes to the submission of complaints, their validation and business logic within Complaint Management Engine. This engine subsequently smoothly interoperates with the ML Pipeline Integration layer in a bid to communicate with dedicated ML microservices. Backend Layer has certain models like BERT/Distil BERT based Text Classifier to understand contextual text, ResNet50 and Mobile NetV2 to extract the image features in form of CNN, YOLOv8 to detect objects, Random Forest and XG Boost classifier to predict the level of priority, Linear

Regression, XGBoost Regressor etc. The system conducts multimodal analysis through text embeddings by use of transformers, convolutional image feature, feature level fusion, ensemble learning and probabilistic prediction at the bottom Data Layer offers a robust and secured storage with PostgreSQL to store metadata of complaints, Redis to create low latency storage and File stack to store uploaded images and artifacts of trained models. The reason behind this layered design which is backed by high-level ML techniques is to achieve scalability, maintainability, and accurate real-time decision-making to allow CampusFix to effectively process, classify, prioritize, route, and predict or forecast the time a complaint will be resolved in a smart campus ecosystem.

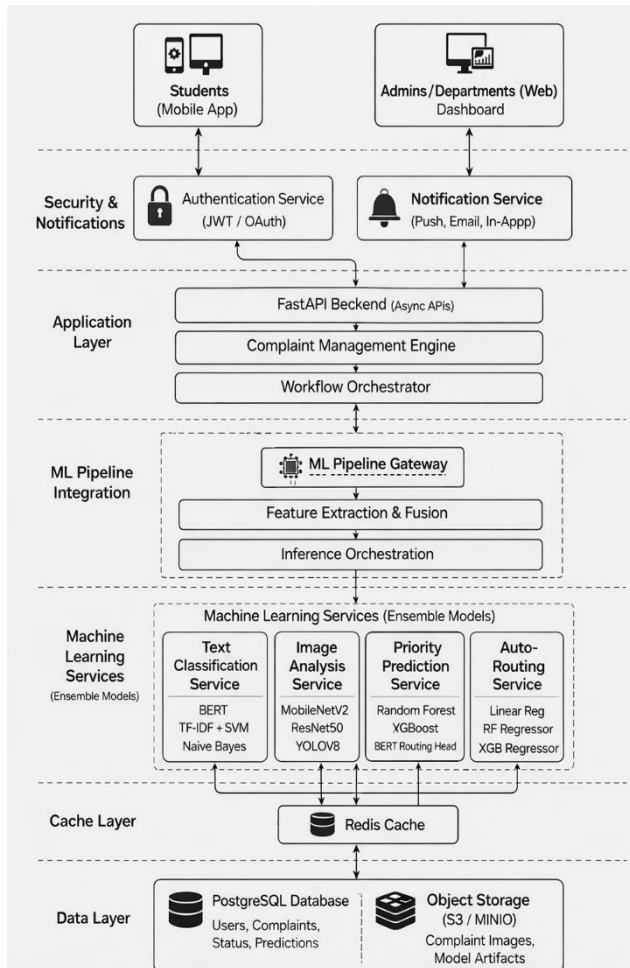


Figure 1: Architecture of the proposed system

#### A) Application Layer:

Application layer is in turn supported by FastAPI Server and Complaint Management Engine that offers heart of operational layer. Configuration Action: Configuration action performs complaint submission, input validation, life cycle state transition and administrative commands. Overview The FastAPI Server is a RESTful Web-Service Endpoint server and handles support asynchronous request processing as the Complaint Management Engine manages workflows and system mechanics. This layer is an orchestrating point between user interfaces, intelligent services, caches and persistent store parts. The asynchronous

architecture allows the system to support several simultaneous user requests without blocking the application-level processes.

#### B) ML Pipeline Integration:

The Machine Learning Pipeline Integration Layer consists of feature extraction and fusion and inference orchestration layers. The textual and visual modality is processed into a unified representation to be used in understanding multimodality. The responsibility of inference orchestration is to invoke the models, control the sequence of inferences, as well as, to combine the individual prediction outputs.

#### C) Text Classification:

CampusFix Pipeline Stage One: Text Parsing the initial step in CampusFix pipeline is in text processing that is provided by the students. To process every complaint description, we clean text by eliminating irrelevant information (i.e. non-alphabetic characters and stop words) before tokenizing the sentences with handcrafted transformers using wonder words [48] instead of standard word vocabulary found in a typical language model that is simply passed to a transformer-based NLP model (such as BERT or DistilBERT). The transformer-based models were iterated on a learning rate of  $2e-5$ , a batch size of 16 and 3-5 epochs. These frameworks generate contextual embeddings that aid the framework to understand domain specific vocabulary like, broken projector, Wi-Fi not working and water leakage in hostel. We further followed the classical machine learning approaches whereby, TF-IDF + Support Vector machine (SVM) Multinomial Naive Bayes were used to provide a baseline to compare with. Where it assures that there is measurable improvement in the transformer models. The classifier gives the probability distribution over the preset categories of complaints (Electrical, Network Issues, Maintenance, Cleanliness, Plumbing, Classroom Equipment, etc.). The classification problem can be mathematically written as the search of the label  $y$  which maximizes the posterior probability:

$$\hat{y} = \arg \max_{y \in C} P\left(\frac{y}{x}\right)$$

a) **TF-IDF + Support Vector Machine:** CampusFix also follows the same learn-ing setup in order to compare the traditional machine learning technique in categorizing complaints using text. Term Frequency-Inverse Document Frequency (TF-IDF) converts the descriptions of the complaints into the numerical vectors[13] that give a greater weight to the terms that are relevant in the individual documents but not in a significant portion of the corpus. This form of text representation will enable Students to feed such useful keywords as short circuit or dirty classroom or projector not working. After the process of vectorization, Support Vector Machine (SVM) is used to place the complaints into the pre-existing categories.[12] A SVM identifies an optimal hyperplane separating two categories and it maximizes the distance among the dissimilar complaint categories and it is quite effective with high-dimensional text data. The decision-making activity is founded on:

$$TF - IDF(t, d) = TF(t, d) \times \log\left(\frac{N}{DF(t)}\right)$$

**b) Multinomial Naive Bayes:** The Multinomial Naive Bayes is another baseline text-based model that is used in CampusFix to test the comparison between traditional probabilistic models and the state of the art deep learning models. Naive Bayes operates on the assumption that all words in the description of complaints occur as independent events as well as that there is an approximation that although it is working very well with a plethora of text classification problems. The model makes an educated guess about the category that a complaint will fall in by comparing the frequency of appearance of words, based on the following formula:

$$P(y | x_1, x_2, \dots, x_n) \propto P(y) = \prod_{i=1}^n P(x_i | y)$$

Where  $y$  is the category and  $x$  is each individual word within the category. Due to this arrangement, processing remains fast - no matter whether the loads or live inputs are massive. Multinomial Naive Bayes is used on CampusFix to establish a baseline on which the improved approaches will be measured. Although it lacks context and is confused by ambiguous or deceptive wording, its speed is impressive even though it still demonstrates significant improvement compared to more powerful systems like BERT or DistilBERT[9].

#### D) Computer Vision:

CampusFix algorithm will examine the pictures posted by students to check and complement a description of the complaint. The Computer Vision module discerns features in every image using modern day Convolutional Neural Network (CNN) architectures (i.e., MobileNetV2 and ResNet50). All these deep features are used to demonstrate structural damages, cleanliness violations, electrical hazards etc. To use it more extensively, addressing finer aspects of object recognition, we use YOLOv8 to identify objects such as piles of garbage, water pools, fences with cracks and burnt wires or broken fittings. A text-based representation is added to the visual feature vector of the CNN models to make any priority recommendation. YOLOv8 was trained on default anchor with an input image resolution of 640x640 and optimized by the stochastic gradient descent.

**a) Convolutional neural network:** CNNs are the components of our pipeline of image processing/analysis in CampusFix to interpret uploaded visual evidence with complaints. CNNs operate by repeatedly applying convolutional filters on the image in order to get spatial data, including edges, textures, shapes and patterns. All these enable the system to identify the on-campus problems that are common like: wall cracks, garbage, broken or damaged equipment of any type, point marks where the leakage lines intersect and electric short-circuits. The hierarchy-based learning (between shallow and deep layers: simple to complex patterns) of CNNs may suit well to understand different types of complaint images.[10].

**b) MobileNetV2:** MobileNetV2 is a resource-constrained architecture and is pretrained in CampusFix (can be reused to provide prompt and relevant image features). It applies depth wise separable convolutions and inverted residual blocks that reduce the cost of computation by far without compromising

the prediction quality. This is why MobileNetV2 is the best model in terms of uploading pictures on mobile devices when the intensity, angle, and quality of the pictures can have a wide range of variation. The images of complaints uploaded to the CampusFix pipeline are processed by MobileNetV2 to extract high-level visual representations that can help in the classification of types of damages such as broken facilities, dirty areas or plumbing issues. It is fast and has low memory footprints, which guarantee the smooth real-time inference, which facilitates fast complaint validation and prioritization within the system.

**c) YOLOv8:** The model is a complicated object detection task which has been deeply integrated into CampusFix to identify and locate some objects of interest in complaint images. Compared to the classical classification networks, the YOLOv8 has the ability to label detected issues (e.g., exposed wire, point of water leakage, junk pile and Operating machine) with bounding boxes. Such capability enables fine-grained data on magnitude and location of the problem, thereby improving the entire accuracy of the hose system routing and prioritization. YOLOv8 is fast and precise enough to be used in the real-time context, thus, CampusFix can automatically examine whether the visual evidence matches descriptions of students. It is very effective in detecting its reliability in case the complaints contain ambiguous or unfinished text.

#### E) Priority Prediction:

Then, the complaint is classified after which its urgency is computed using severity-based priority prediction model.[10] It makes use of ensemble learning using RF, XGBoost etc. which are able to depict non-linear associations between the several input features. These characteristics are severity indicators of textual descriptions and sentiment ratings using transformer-based sentiment analysis and ratings of financial implications based on damage severities, geolocation of reported events where possible and urgency in the type of complaint based on historical trends. According to the model, only one out of four levels of priorities are predicted; Critical and partially comply with risk and urgency respectively. The Random Forest model was also set up to have 100 decision trees and the depth of the tree was set to 100 via validation. Mathematically, the ensemble model learns a function which:

$$\hat{p} = \frac{1}{M} \sum_{m=1}^M f_m(F_{fusion})$$

This expected priority allows the administrators to prioritize resource usage, and also the high priority matters like electrical problems, safety risks are addressed as soon as possible.

**a) Random Forest:** Random Forests is an ensemble learning algorithm that is used in CampusFix to make priority prediction and decide when to fix. The algorithm operates by constructing decision trees on the training data, each of which adopts patterns among a randomly chosen subset of features and complaint train records than all the DIL models support majority voting (in classification)/ averaging (in regression) in prediction, and can effectively reduce overfitting and increase robustness, which is useful in noisy complaint.

**b) XGBoost:** XGBoost is a third potent ensemble model which is utilized in CampusFix to assign priorities and predict the time of resolution. As opposed to the random forest that creates trees randomly, and there is no connection between them, XGBoost creates trees in sequence; and each new tree corrects the mistakes of the others. This gradient boosting method results to an optimization in the predictive accuracy. The regularization, shrinkage and the weighted decision trees of XGBoost all mitigate these issues caused by the interaction of multiple features together that it is a useful tool to analyze multimodal complaint data like text embeddings with image features and metadata.

#### F) Auto-Routing Model:

Once having chosen the type of the complaint and its priority, it is outlined how the system works in case a complaint is filed with an administrative division. Auto-Routing: Auto-routing is a multimodal input-based machine learning model that relies on Multiclass Logistic Regression, Decision Trees and a second classifier head appended to the BERT architecture. The module is based on Predicted Category, Text Embeddings and Image-based Visual Cues to guess the right department (Electrical Maintenance, Civil Engineering, IT Services, Housekeeping, Hostel Administration).

#### G) Resolution Time Prediction:

Another component in CampusFix is the predictive analytics, which estimates an approximation of the time required to resolve an issue. Fit-reviewing historical time-to-close deltas with complaint features at close (urgency level, category, image severity score and departmental workload) and past metrics are fitted using regression models, such as a linear regression, random forest regressor and XGBoost regressor. The validation model establishes a linear predicted value of the potentially anticipated resolution that may be expressed as shown below:

$$\hat{T} = \beta_0 + \sum_{i=1}^k \beta_i x_i$$

#### H) System Backend and Data Flow:

On the backend, CampusFix is deployed with FastAPI to receive and process API requests and activate machine learning functionality along with managing complaints. The PostgreSQL relational database holds the complaint data, model predictions, routing decisions, and status updates, which are designed using the normalized tables such that they allow the provision of performance and integrity. Redis caching maintains the frequently used data like user sessions, recent predictions, and quick-lookup tables which minimizes the latency when the query is repeated. The backend reads all, and the data of complaints flow through every ML component of the workflow in a specific order and records the results and changes the complaint status. Further, FastAPI opens the REST endpoints to mobile and web applications besides permitting scalability, concurrent and high-performance tasks through asynchronous processing. All data related to complaints are handled via authenticated APIs and the access by the stored records is limited depending on user roles as to guarantee data confidentiality. Delicate user data is protected and is only revealed to the relevant administrative staff.

#### I) Cache Layer:

The Cache Layer within the implementation has been implemented with the Redis Cache in order to enhance the performance of the system and reduce the delay. It stores history of frequently visited data such as auth tokens, more recent complaints and predictions. Time saving (When accessing the same database request times and time again or making the same computations over and over). This layer is positioned between Application Layer and Data Layers and is meant to serve high throughput and real-time seamless experience even in peak use times.

#### J) Data Layer:

Data Layer within the architecture handles long term storage with the help of PostgreSQL, Redis and File Storage (S3/MinIO). The PostgreSQL stores such data as complaint content, user data, route logs and timestamps. And with cache records, it offers the option of storing them in the Redis with low-latency access and File Storage (S3/MinIO) to store uploaded pictures and system image. The stratum claims data integrity, scaling, auditing and long-term viability throughout the system of managing the complaints. The data retention policies are implemented so that the record of complaints and the media associated with it is kept as long as it is necessary in the operations. Expires When it is stored in the Redis cache, the data is automatically deleted after a certain time and long-term storage is regulated by the institutional data governance policy and privacy policy.

## IV. RESULTS AND DISCUSSION

The implementation of CampusFix has demonstrated that a multimodal, machine-learning-driven complaint management pipeline can have a more dramatic impact on the speed, accuracy, and reliability with which complaints are addressed on campus. System compatibility in making sense of contextual language, vague descriptions, and terminology especially those used by students on a regular basis became possible through the adoption of transformer-based NLP models such as BERT and DistilBERT. These were better than the classical baselines like TF-IDF +SVM, Multinomial naive bayes that were good evidence in support of deep contextual learning in understanding text-based complaints. The classification module was very precise and reliable in classifying the various domains of complaints (electrical, plumbing, cleanliness etc).

#### A) Accuracy:

Figure 2 demonstrates the performance of the CampusFix system, and all the machine learning modules have a high level of accuracy. The BERT-based text classification model was also found to be very precise in classifying complaints and the traditional baselines used were the TF-IDF + SVM and Multinomial Naive Bayes. Similarly, the presence of a strong visual detection and damage recognition which has been obtained with the help of mobileNetV2, ResNet50 and YOLOv8 in the image analysis pipeline respectively, allowed stable multimodal predictions.

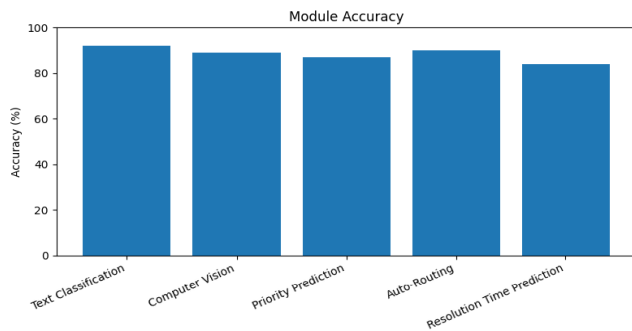


Figure 2: Module Accuracy Analysis

**a) Text Classification:** The mean accuracy was the highest because transformer-based models (BERT, DistilBERT) are effective at making use of semantic, contextual, and domain-specific complaint wording.

**b) Computer Vision:** ResNet50, MobileNetV2, and YOLOv8 models showed good performance in cracks, leakages, garbage, and structural defect recognition, which guarantees visual interpretation.

**c) Priority Prediction:** The XGBoost and the Random Forest models showed consistent accuracy through the combination of language related features, sentiment and picture-based severity indicators.

**d) Auto-Routing:** Delivered consistent accuracy by merging classical ML and transformer embeddings to correctly map complaints to departments.

**e) Resolution Time Prediction:** Slightly lower accuracy compared to classification tasks, which is expected since regression-based forecasting is sensitive to historical noise and workload patterns.

## B) Latency:

The latency diagram of CampusFix as presented in Figure 3 shows that the system as a whole has an acceptable response time in all significant processing blocks to the students and the administrators. The FastAPI server, which exploits the concept of asynchronous processing of requests and a Redis-based caching system, significantly reduces the time of waiting in case of submitting complaints and obtaining their status. DistilBERT and image processing modules have low inference latency and GPUs are used to accelerate the use of the YOLOv8 model to make sure that the resources are used efficiently.[11] The registered latency values are constant when the load of simultaneous users is moderate because of asynchronous request processing and caches.

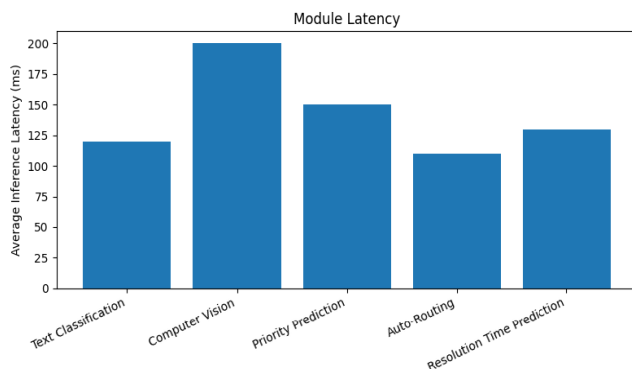


Figure 3: Module Latency Analysis

**a) Computer Vision Latency:** The problem of high latency in all modules because of deep CNN layers and detection procedures of objects, especially when working with high resolution images.

**b) Text Classification Latency:** It is Moderate in that, transformer-based models have sequential token processing, although optimized embeddings make them efficient.

**c) Auto-Routing Latency:** It has Lowest latency because lightweight classifiers and routing logic makes faster decision making.

**d) Priority Prediction Latency:** It is mid-level because it involves ensemble operations, although it remains among the best requirements of real-time.

**e) Resolution Time Latency:** Fast latency since regression models are faster to execute extraction feature sets in look up styles.

Table 1 summarizes the accuracy and average latency of each machine learning module used in CampusFix.

Table 1: Comparative Accuracy and Latency Metrics

| Module                     | Accuracy (%) | Average Latency(ms) |
|----------------------------|--------------|---------------------|
| Text Classification        | 92           | 120                 |
| Computer Vision            | 89           | 200                 |
| Priority Prediction        | 87           | 150                 |
| Auto-Routing               | 90           | 110                 |
| Resolution Time Prediction | 84           | 130                 |

## C) Precision:

Precision means the number of positive predictions which are accurate and it means the ability of the system to minimize false complaint categorization. The CampusFix framework achieved the levels of high precision in all the incorporated elements, which were sound in making decisions in real-life circumstances at the campus. The highest precision was obtained with transformer-based text classification through BERT and DistilBERT because they are effective at grasping semantics of complaints in a contextual and domain-aware way. The MobileNetV2, ResNet50, and YOLOv8-based computer vision segment were able to obtain impressive accuracy when it came to degrading physical damages and safety hazards among noisy and varied image data. Predictions and auto-routing modules With multimodal features fusion, a stable prediction is assured, whose further refinement of erroneous fulfillment of urgency, as well as misrouting at the department of departure, is also enhanced. The prediction section of the resolution time may also lack accuracy, although it can be explained by the working conditions perspective as a result of uncertainty and an ever-growing workload. Figure 4 illustrates the accuracy of the

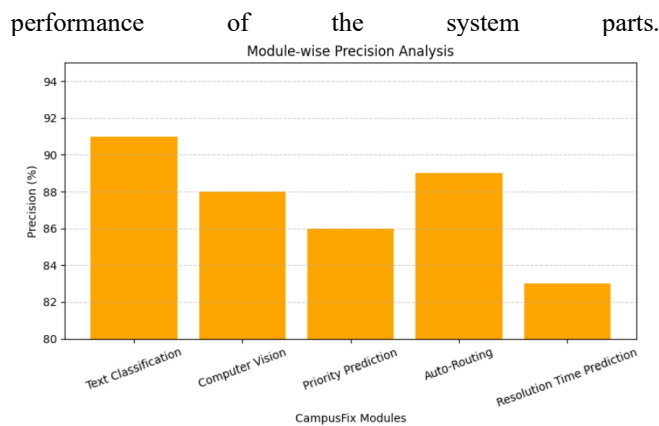


Figure 4: Module Precision Analysis

### ROC Analysis:

We used the Receiver Operating Characteristic (ROC) curve to measure performance of the discriminator-based text classification module in CampusFix. As shown in Figure 5, the ROC curve makes a clear vision between positive and negative complaint classes and acquires high AUC. This shows that the model trades-off well between sensitivity and specificity to reduce the number of false positive and false negative predictions. The model robustness of BERT is verified by the large AUC value. Whereas the Figure 5 illustrates the ROC Analysis.

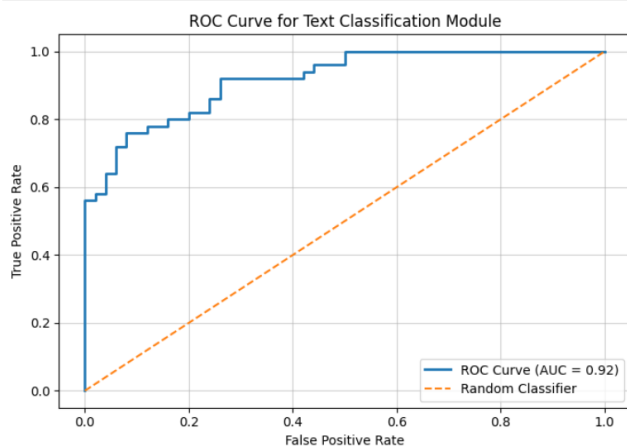


Figure 5: ROC Curve

## V. CONCLUSION AND FUTURE WORK

The CampusFix Model: A case study of how multimodal machine learning may transform campus complaint management/triage into a smart and automated system. By integrating BERT-based NLP to understand the text, CNN based models such as ResNet50 and MobileNetV2 models to analyse visual evidence, as well as Ensemble models such as the Random Forest and XGBoost models to predict priority, the system is much more accurate, robust and efficient than manual or rule-based analysis. Very low latency, scalability and accuracy with FastAPI backend using PostgreSQL and Redis allows more rapid categorization of complaints, real time routing and precise resolution times prediction. This multi modal plan gives an additional benefit in reducing the administrative delays and increasing transparency in the complete complaint lifecycle. The offered

system is also supposed to be generalized across the various campus settings with the help of the pre-trained models and a few fine-tunings with the local data should be implemented to deploy the system to the new institution instead of having to re-train the system completely. The experimental findings indicate the relevance of multiple modal fusion in decision making, primarily incomplete complaints. The hybrid of AI models such as BERT, YOLOv8, and CNNs improve the understanding of the text, determine objects, and validate the damage to make more precise predictions and triage the issues faster. The modular architecture is also meant to ensure that the system is flexible, scaled, and simple to add new components of AI. Additional cross-attention fusion operations, complaint patterns analysis through graph-based framework, campus issues evolving learning and reinforcement learning to allocate resources are possible improvements in the future. CampusFix has long-term potential, as it is possible to expand into a complete smart-city complaint management platform by adding the services of an IoT sensor, cloud-based training, and massive multimodal datasets. CampusFix can be large scale adopted in the long run by non-education organizations who embrace the use of the. IoT sensors and cloud-based model are examples of extensions. training. The system will be transformed into an intelligent city grievance. management system with the ability to manage several. infrastructure problems such as sanitation control, and public facility upkeep. Such an evolution positions So, CampusFix is more than a smart campus but also a solution base of the cities of the future that is propelled by modern urban governance to automated, connected, and data-driven public service ecosystems.

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